

EDUCATION

PhD in Chemistry and Chemical Engineering | EPFL, Swiss Federal Institute of Technology, Lausanne 2023 – Present

Supervisors: Prof. Philippe Schwaller, Dr. Raphael Bigler, Dr. Kurt Püntener

- Jointly hosted by Roche (Basel) and the Laboratory of Artificial Chemical Intelligence (EPFL)

Visiting Doctoral Student | ETH Zürich, Swiss Federal Institute of Technology, Zürich 2024 – Present

- Courses: Science to Medicines (6/6), Digital Chemistry (5.75/6), Computer-Aided Drug Design (ongoing)

MChem in Physical and Theoretical Chemistry | Balliol College, University of Oxford 2018 – 2023

- Thesis: “Directing Heterocycle Synthesis with Machine Learning” (Supervisor: Prof. Fernanda Duarte)
- First Class Honors, Distinction in all public examinations and in thesis

RESEARCH EXPERIENCE

PhD Student | Roche, Synthetic Molecules Technical Development Oct 2023 – Present

- Designed workflows using Bayesian optimization, graph neural networks, and large language models for reaction prediction and optimization across early-stage drug discovery, medicinal and preclinical chemistry, and process chemistry teams; resulting in three (co-)first and co-corresponding author publications
- Secured competitive project funding for two internship students; designed research proposals and supervised projects spanning biocatalysis, metaheuristic algorithms, and generative modeling for reaction condition prediction
- Executed high-throughput experimentation (HTE) campaigns end-to-end, designing reaction plates, writing automated solid dosing and robotic protocol workflows, and running experiments on integrated robotic platforms
- Developing agentic systems with long-term memory for reaction optimization
- Invited to review manuscripts for *Chem* (Cell Press), *Nature Communications* (Nature portfolio), *npj Computational Materials* (Nature portfolio), *Chemical Science* (RSC), *Digital Discovery* (RSC)

Research Assistant | University of Oxford, Physical and Theoretical Chemistry Laboratory Sep 2022 – Aug 2023

Supervisor: Prof. Fernanda Duarte

- Developed Computer-Aided Synthesis Planning (CASP) retrosynthetic tools for accelerating heterocycle bioisostere synthesis in drug discovery (contributing author, published in *Journal of Chemical Information and Modeling*)
- Applied transfer learning and ensemble decoding methods to optimize transformer architectures for heterocyclic retrosynthesis
- Integrated single-step retrosynthesis tools with tree search algorithms including Monte Carlo Tree Search and Retro* for benchmarking multi-step retrosynthesis of drug-like molecules
- Built cheminformatic workflows to extract reaction rules and templates from literature databases, using transformer models to annotate and classify reaction data

Research Scholar | Agency for Science, Technology and Research (A*STAR) Singapore Jun 2022 – Aug 2022

Supervisors: Dr. Yee Hwee Lim, Dr. Dillon Tay

- Developed machine learning models to optimize enzyme stability and activity in reverse micelle systems for renewable biocatalysis
- Applied cheminformatic fingerprints, physical chemistry descriptors, and learned graph neural network representations to predict kinetic activity of enzymatic systems

PUBLICATIONS

*Corresponding author; †Equal contribution.

1. Dynamic language model representations for multi-objective reaction optimisation |

Preprint, under review – earlier version accepted at ICML 2026, AI for Science Workshop

[code](#) [pdf](#)Joshua W. Sin[†], David Ming Segura[†], Bojana Ranković[†], Siu Lun Chau, Marius D. R. Lutz, Andrea Anelli, Ryan P. Burwood, Kurt Püntener, Maximilian J. Notheis, Raphael Bigler, Philippe Schwaller*

2. Reaction: Navigating the path to reasoning laboratories |

Chem (Cell Press), **12**, 8 (2026)[pdf](#)Joshua W. Sin^{*†}, David Ming Segura[†], Ramon Winkler[†], Philippe Schwaller*

3. Bi-semantic Chemical Embedder for Joint Representation Learning of SMILES and Natural Language |

Preprint, under review

code

pdf

David Ming Segura[†], Jeremy Goumaz[†], **Joshua W. Sin**, Bojana Ranković, Philippe Schwaller*

4. Swarm intelligence for chemical reaction optimization |

Chem (Cell Press), **12**, 103035 (2026)

code

pdf

Rémi Schlama[†], **Joshua W. Sin**[†], Ryan P. Burwood, Kurt Püntener, Raphael Bigler, Philippe Schwaller*

5. Highly parallel optimisation of chemical reactions through automation and machine intelligence |

Nature Communications, **16**, 6464 (2025)

code

pdf

Joshua W. Sin*, Siu Lun Chau, Ryan P. Burwood, Kurt Püntener, Raphael Bigler, Philippe Schwaller*

6. Transfer Learning for Heterocycle Retrosynthesis |

Journal of Chemical Information and Modeling, **65**, 15 (2025)

code

pdf

Ewa Wieczorek, **Joshua W. Sin**, Sara Tanovic, Matthew T. O. Holland, Liam Wilbraham, Victor S. Perez, Anthony Bradley, Dominik Miketa, Paul E. Brennan, Fernanda Duarte*

RESEARCH FUNDING AND AWARDS

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| • Technology Innovation and Science Initiative Doctoral Studentship, Roche | 2023 |
| • Singapore International Pre-Graduate Award (SIPGA), A*STAR Singapore | 2022 |
| • Andrew Pang Scholarship, Balliol College, University of Oxford | 2021 |
| • Brackenbury Exhibition, Balliol College, University of Oxford | 2019 |

INVITED TALKS AND PRESENTATIONS

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| 'Optimizing Chemical Process Development with Automation and Machine Learning', University of Zürich | May 2026 |
| 'Digitalization and Automation in Chemistry', University of Applied Sciences & Arts Northwestern Switzerland | Sep 2025 |
| 'Modeling and Feature Representations in Closed Loop Discovery', Bayer Closed Loop Consortium, Wuppertal | May 2025 |
| 'Combining HTE and Data Science to Optimize Chemical Processes', Automated Synthesis Forum, Oxford | Nov 2024 |

SUPERVISION EXPERIENCE

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| Ramon Winkler , Technical R&D Senior Associate Scientist at Roche | Aug 2025 – Present |
| David Ming Segura , Intern at Roche | Jul 2025 – Dec 2025 |
| Lorenzo Tarricone , Master's student at ETH Zürich | Mar 2025 – Aug 2025 |
| Elio Gianinazzi , Intern at Roche | Jan 2025 – Sep 2025 |
| Rémi Schlama , Master's student at EPFL | Jul 2024 – Nov 2024 |

SKILLS

Programming: Python, Git, SLURM | **ML**: PyTorch, PyTorch Geometric, Hugging Face, scikit-learn, GPyTorch, BoTorch, Weights & Biases | **Chemistry**: RDKit, High-throughput Experimentation (HTE)